

Tectonic Lithography: A Novel True Random Number Generator Based on Seismic Cross-Correlation

Reese Wilson
Manifold Dynamics (Praetor Defense)

December 2025

Abstract

We present Tectonic Lithography, a novel true random number generator (TRNG) that derives cryptographic entropy from the cross-correlation of geographically separated seismometer stations. By exploiting the decorrelated microseismic noise between distant sensors monitoring Earth’s continuous seismic background, we generate high-quality random numbers that are physically tied to planetary-scale processes. Our implementation achieves entropy extraction through SHA-256 hashing of cross-correlation waveforms, producing uniformly distributed 256-bit keys. The system demonstrates robustness through automatic station failover across 545+ candidate seismometers and targets the secondary microseism frequency band (0.1-0.8 Hz) where oceanic interference patterns dominate. We discuss the theoretical entropy sources, implementation architecture, security properties, and potential applications of this geophysically-grounded approach to random number generation.

1 Introduction

True random number generators (TRNGs) are fundamental to cryptographic systems, secure communications, and scientific computing. Unlike pseudorandom number generators (PRNGs) which rely on deterministic algorithms, TRNGs extract entropy from physical processes that are inherently unpredictable [1].

Traditional TRNG entropy sources include thermal noise in semiconductors, quantum phenomena such as photon arrival times, radioactive decay, and atmospheric noise [2]. Each approach offers distinct advantages in terms of entropy rate, implementation complexity, and verification methods.

This work introduces a fundamentally different entropy source: *the differential seismic noise between geographically separated seismometer stations*. We term this approach “Tectonic Lithography,” reflecting its reliance on Earth’s lithospheric dynamics as the physical substrate for randomness generation.

1.1 Motivation

The Earth’s surface is in continuous motion, experiencing microseismic activity from numerous sources: ocean wave interference (microseisms), atmospheric pressure variations, thermal expansion, tectonic stress, and anthropogenic activity. Modern seismometer networks provide real-time access to this data through services such as IRIS (Incorporated Research Institutions for Seismology).

While individual seismic stations record coherent signals from large events (earthquakes, storms), the high-frequency noise floor between distant stations is largely decorrelated due to:

- Localized cultural noise (traffic, machinery, construction)
- Regional geological structure variations
- Differential atmospheric coupling
- Phase dispersion across propagation paths
- Independent sensor electronics noise

This decorrelation creates a natural entropy source that is:

1. **Globally accessible:** Any party can retrieve the same raw seismic data
2. **Locally unpredictable:** Cross-correlation phase relationships are chaotic
3. **Historically verifiable:** Data is archived and timestamped
4. **Physically grounded:** Tied to real planetary processes, not electronic artifacts

2 Theoretical Framework

2.1 Entropy Sources in Seismic Noise

The seismic noise spectrum can be decomposed into several components:

$$S(f) = S_{\text{microseism}}(f) + S_{\text{cultural}}(f) + S_{\text{instrument}}(f) + S_{\text{thermal}}(f) \quad (1)$$

where f represents frequency. For our purposes, we focus on the *secondary microseism* band (0.1-0.8 Hz), where ocean wave interference dominates [3].

2.1.1 Secondary Microseisms

Ocean waves traveling in opposite directions create standing wave patterns that couple to the seafloor, generating seismic waves at twice the ocean wave frequency. This phenomenon produces a globally coherent signal in the 0.1-0.8 Hz band, but with complex spatial and temporal variations due to:

- Storm system locations and intensities
- Coastal bathymetry and geometry
- Wave-wave interaction nonlinearities
- Seasonal ocean condition variations

2.2 Cross-Correlation as Entropy Extraction

Given two seismic traces $x_1(t)$ and $x_2(t)$ from stations separated by distance d , the cross-correlation function is:

$$R_{12}(\tau) = \int_{-\infty}^{\infty} x_1(t) \cdot x_2(t - \tau) dt \quad (2)$$

For our implementation, we compute the discrete cross-correlation over a finite window $T = 600$ seconds:

$$R_{12}[k] = \sum_{n=0}^{N-1} x_1[n] \cdot x_2[n - k] \quad (3)$$

where N is the number of samples and k represents the lag index.

2.2.1 Synchronization Score

We define a synchronization score as:

$$\sigma = \max_k |R_{12}[k]| \quad (4)$$

This metric quantifies the degree of coherent signal between stations. Empirically, we observe $\sigma \approx 0.2 - 0.4$ for station pairs separated by ~ 1500 km, indicating that 60-80% of the signal is decorrelated noise.

3 Implementation

3.1 System Architecture

The Tectonic Lithography system consists of four primary modules:

1. **Geo-Locator:** Determines user position via IP geolocation
2. **Station Discovery:** Queries IRIS FDSN web services for nearby high-resolution (BHZ channel) seismometers
3. **Data Acquisition:** Retrieves time-synchronized seismic traces with automatic failover
4. **Entropy Extraction:** Processes cross-correlation waveforms through cryptographic hashing

3.2 Station Selection Protocol

The system implements an automatic failover mechanism across multiple seismic networks (US, IU, GS, TA, N4, CI, CO). For a given geographic location (lat, lon), candidate stations are discovered within an 8-degree radius (≈ 800 km).

The reference station is fixed as IU.ANMO (Albuquerque, New Mexico), one of the most reliably operational stations in the Global Seismographic Network.

Algorithm 1 Station Failover Protocol

```
1:  $candidates \leftarrow \text{QueryStations}(lat, lon, radius = 8)$ 
2: for each  $(network, station)$  in  $candidates$  do
3:    $trace \leftarrow \text{FetchData}(network, station, t_0, duration)$ 
4:   if  $trace \neq \text{null}$  then
5:     return  $trace$ 
6:   end if
7: end for
8: return FAILURE
```

3.3 Signal Processing Pipeline

Each retrieved seismic trace undergoes the following preprocessing:

1. **Detrending:** Linear trend removal to eliminate thermal drift
2. **Tapering:** 5% cosine taper at trace edges to reduce spectral leakage
3. **Bandpass Filtering:** 4th-order Butterworth filter with passband $[0.1, 0.8]$ Hz
4. **Normalization:** Amplitude scaling to $[-1, 1]$ for numerical stability

The filtering stage is critical for isolating the secondary microseism band where entropy density is highest.

3.4 Cryptographic Key Generation

The core innovation of our approach is the application of cryptographic hashing to the cross-correlation waveform. Traditional TRNG designs struggle with non-uniform bit distributions from physical sources. Our solution:

Algorithm 2 Entropy Extraction via Hashing

```
1:  $cc \leftarrow \text{CrossCorrelate}(x_1, x_2, maxlag = 1000)$ 
2:  $center \leftarrow \arg \max_k |cc[k]|$ 
3:  $segment \leftarrow cc[center - 500 : center + 500]$ 
4:  $bytes \leftarrow \text{ToBytes}(segment)$ 
5:  $key \leftarrow \text{SHA256}(bytes)$ 
6: return  $key$ 
```

By extracting a 1000-sample window centered on the peak correlation and hashing the raw floating-point representation, we achieve:

- **Uniform distribution:** SHA-256 provides cryptographic whitening
- **Avalanche effect:** Microscopic waveform differences produce completely different keys
- **Fixed output length:** 256 bits regardless of input trace length

4 Security Analysis

4.1 Entropy Estimation

The entropy per key depends on the unpredictability of the cross-correlation waveform shape. Given:

- Floating-point precision: 64 bits per sample
- Extracted window: 1000 samples
- Effective degrees of freedom: ~ 500 (accounting for correlation structure)

A conservative lower bound on min-entropy is:

$$H_{\min} \approx 500 \times \log_2(2^{16}) = 8000 \text{ bits} \quad (5)$$

where we assume 16 bits of entropy per decorrelated sample based on the observed $\sigma \approx 0.26$ sync score. SHA-256 compression preserves at least 256 bits of this entropy in the final key.

4.2 Attack Surface

4.2.1 Prediction Attacks

An adversary attempting to predict future keys would need to:

1. Model Earth’s global microseism field in real-time
2. Predict local cultural noise contributions
3. Account for instrument-specific noise characteristics

This is computationally infeasible given the chaotic nature of atmospheric and oceanic dynamics.

4.2.2 Replay Attacks

The system retrieves data from $t_0 = \text{current_time} - 1200$ seconds, ensuring that keys are always based on recent seismic activity. Historical data cannot be replayed since the timestamp is part of the retrieval context.

4.2.3 Infrastructure Attacks

An adversary with control over the IRIS data pipeline could theoretically inject false seismic data. However:

- IRIS serves multiple independent users; data tampering would be detectable
- The cross-correlation operation requires consistent manipulation across two independent stations
- Users can verify key generation by independently downloading and processing archived data

5 Experimental Results

5.1 Key Uniqueness

We generated 1000 consecutive keys at 1-minute intervals. SHA-256 collision analysis confirmed:

- Zero collisions observed
- Hamming distance distribution centered at 128 bits (expected for uniform random keys)
- No detectable autocorrelation in the key sequence

5.2 Synchronization Score Statistics

Over a 24-hour observation period, sync scores for the CCM-ANMO station pair exhibited:

Metric	Value
Mean σ	0.267
Std. Dev.	0.083
Min σ	0.091
Max σ	0.582

Table 1: Synchronization score statistics (n=1440 samples)

Elevated sync scores ($\sigma > 0.4$) correlated with regional seismic events visible in both station records, confirming the physical interpretation of the metric.

6 Discussion

6.1 Comparison to Existing TRNGs

TRNG Type	Entropy Rate	Verifiability	Hardware Cost
Thermal Noise	High	Low	Low
Quantum (Photon)	Very High	Medium	High
Radioactive Decay	Low	High	Medium
Atmospheric Noise	Medium	Medium	Low
Seismic (This Work)	Medium	High	None*

Table 2: Comparison of TRNG approaches (*requires internet access)

Tectonic Lithography occupies a unique niche: it requires no specialized hardware beyond internet connectivity, yet provides physically verifiable entropy tied to planetary-scale processes.

6.2 Limitations

1. **Latency:** 20-minute delay for data availability
2. **Dependency:** Requires operational seismic network infrastructure
3. **Throughput:** Limited to one key per time window (though parallelization across multiple station pairs is possible)

4. **Geographic Variation:** Performance depends on proximity to active seismometer networks

6.3 Future Work

6.3.1 Multi-Station Entropy Pooling

XOR-combining keys from multiple independent station pairs could increase entropy rate:

$$K_{\text{final}} = K_1 \oplus K_2 \oplus \cdots \oplus K_n \quad (6)$$

6.3.2 Real-Time Entropy Assessment

Implementing online statistical tests (e.g., NIST SP 800-90B) would enable dynamic quality monitoring and adaptive sampling strategies.

6.3.3 Distributed Key Generation Protocol

Multiple parties could generate shared keys by agreeing on station pairs and time windows, enabling geographically distributed applications without trusted setup.

7 Applications

7.1 Cryptographic Seeding

High-quality entropy for initializing PRNGs in cryptographic applications where hardware RNGs are unavailable.

7.2 Provably Fair Systems

Gaming and lottery systems can use publicly verifiable seismic data to demonstrate fairness without relying on trusted third parties.

7.3 Scientific Computing

Monte Carlo simulations and statistical modeling requiring high-quality randomness with verifiable provenance.

7.4 Generative Art

Creating artworks seeded by specific seismic events, tying creative output to planetary history.

8 Conclusion

We have demonstrated a novel TRNG architecture based on seismic cross-correlation that leverages Earth’s continuous microseismic activity as an entropy source. The system achieves cryptographically strong randomness through SHA-256 hashing of decorrelated seismic noise between distant stations, with built-in robustness through automatic station failover.

Tectonic Lithography represents a paradigm shift in thinking about entropy sources: rather than miniaturizing noise sources into silicon, we scale up to planetary processes. This approach

offers unique properties including public verifiability, historical reproducibility, and zero hardware cost.

While not suitable for all applications due to latency and throughput constraints, this work opens new possibilities for geophysically-grounded cryptographic systems and demonstrates that the Earth itself can serve as a cryptographic oracle.

Code Availability

The complete implementation is available under open-source license at: [repository URL]

References

- [1] Herrero-Collantes, M., & Garcia-Escartin, J. C. (2017). Quantum random number generators. *Reviews of Modern Physics*, 89(1), 015004.
- [2] Walker, J. (1998). HotBits: Genuine random numbers, generated by radioactive decay. *Fourmilab*, accessible at fourmilab.ch/hotbits
- [3] Longuet-Higgins, M. S. (1950). A theory of the origin of microseisms. *Philosophical Transactions of the Royal Society of London. Series A*, 243(857), 1-35.